

CV 5101

Modeling, Uncertainty, and  
Data for Engineers

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# Flow

- Module Outline
- Motivation
- Introduction

# Why should you care about Risk & Reliability?

- Space Shuttle *Challenger* disaster (1986)

<https://www.youtube.com/watch?v=yibNEcn-4yQ>



**Catch-all approach:**

Build for every “what if” scenario: create backup for every imaginable scenario, even wildly unlikely ones

What could go wrong with this approach?

# Why should you care about Risk & Reliability?

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## Catch-all approach:

Build for every “what if” scenario: create backup for every imaginable scenario, even wildly unlikely ones

**Spreads engineers too thin:** divided resources, real threats do not get the focus they need

**Illusion of safety:** looks thorough on paper, but critical risks can slip through because everything is treated as equally important

Questions, comments,  
or concerns?